FEVER: figures

System design

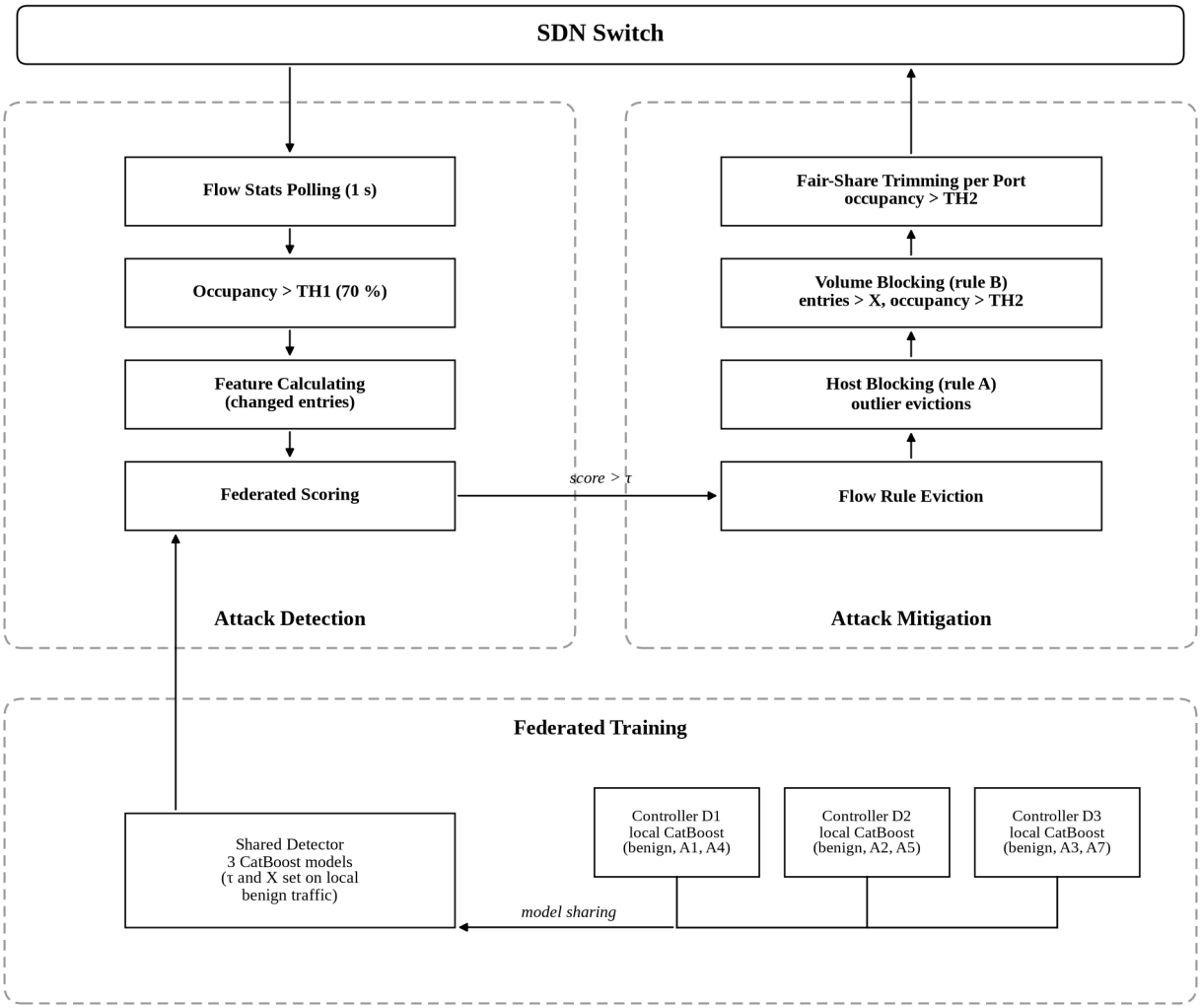


Fig. 1. Architecture of FEVER.

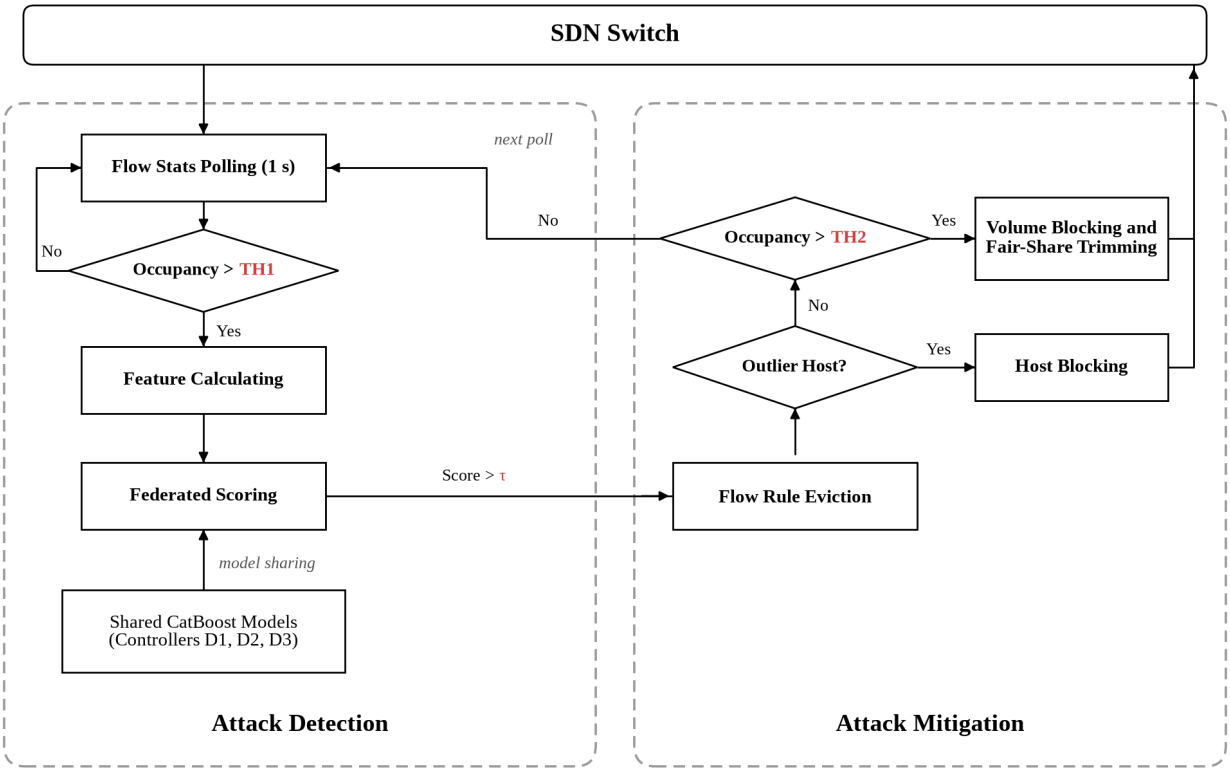


Fig. 2. Framework of FEVER.

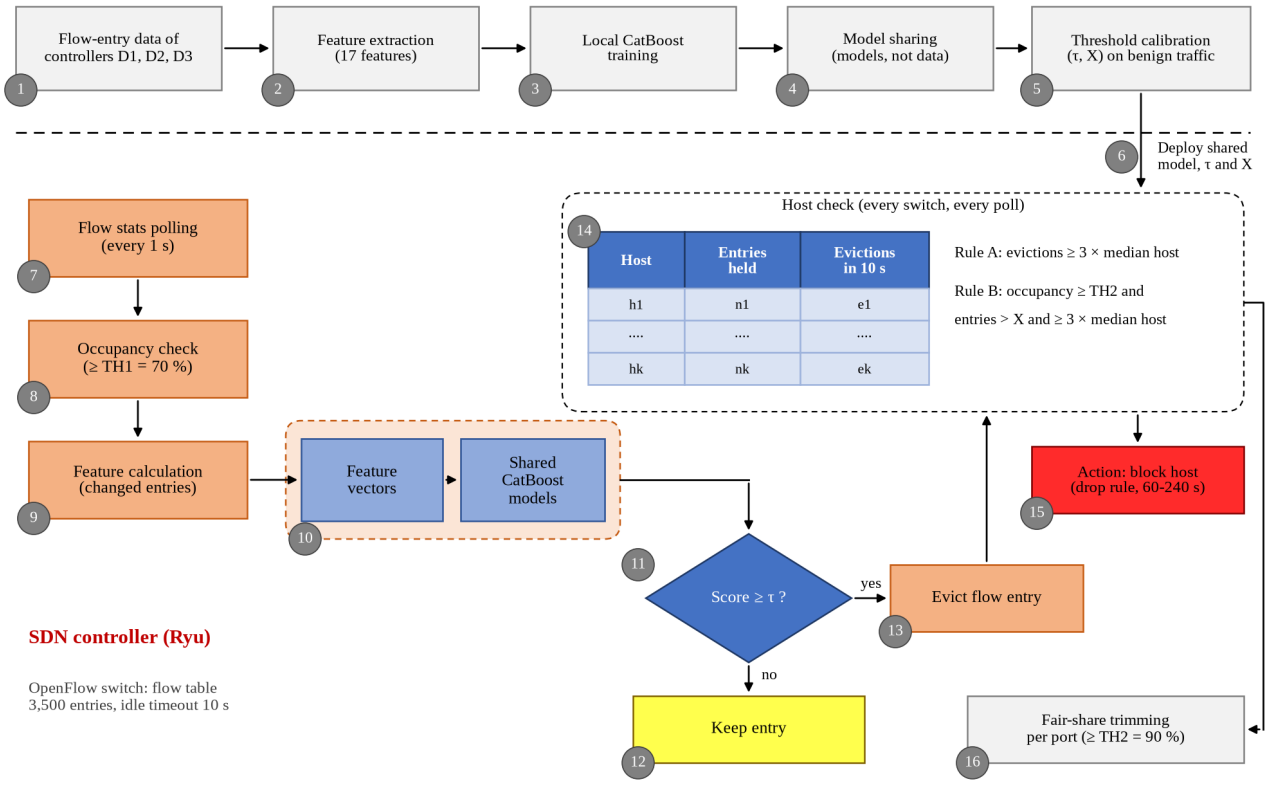


Fig. 3. Working of FEVER.

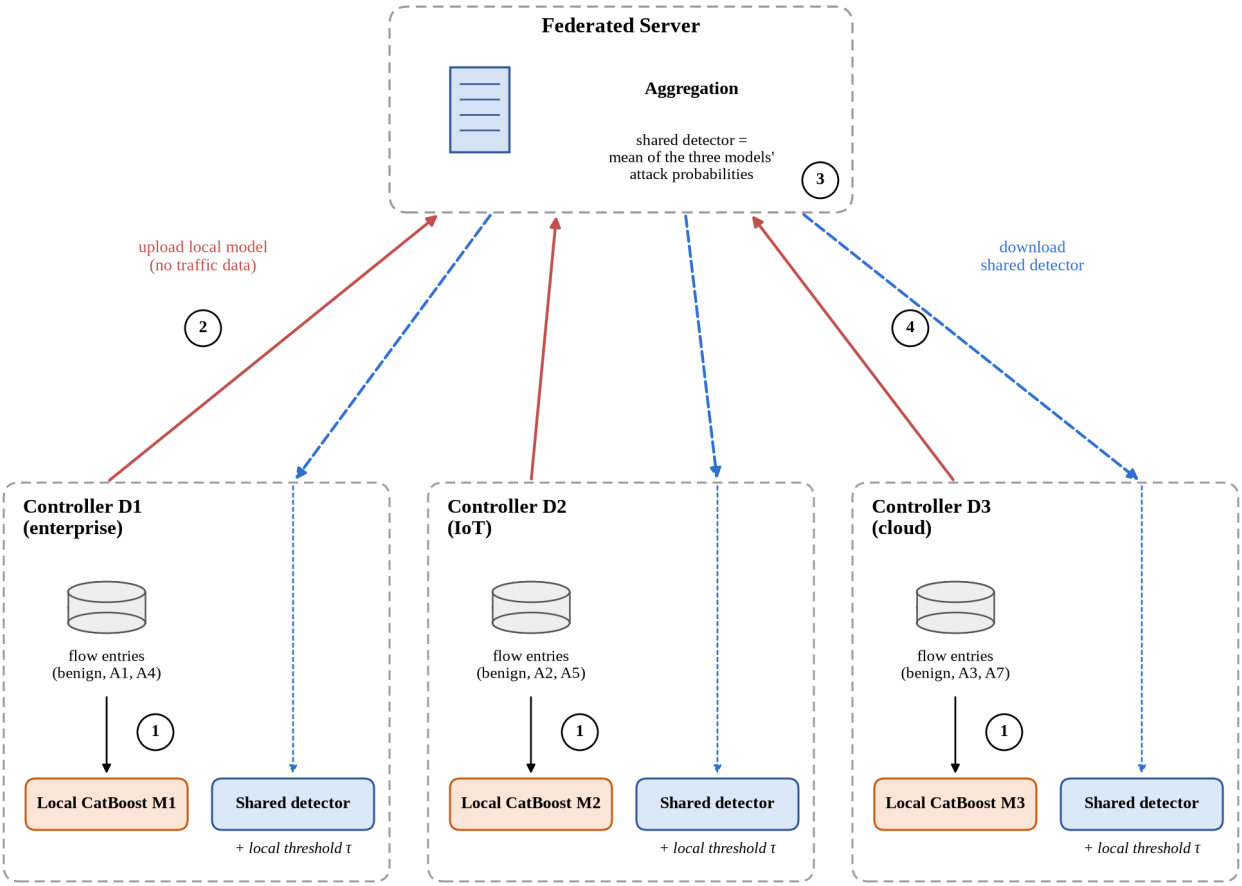


Fig. 4. Federated training of FEVER's detector: (1) local training, (2) model upload, (3) aggregation, (4) download of the shared detector.

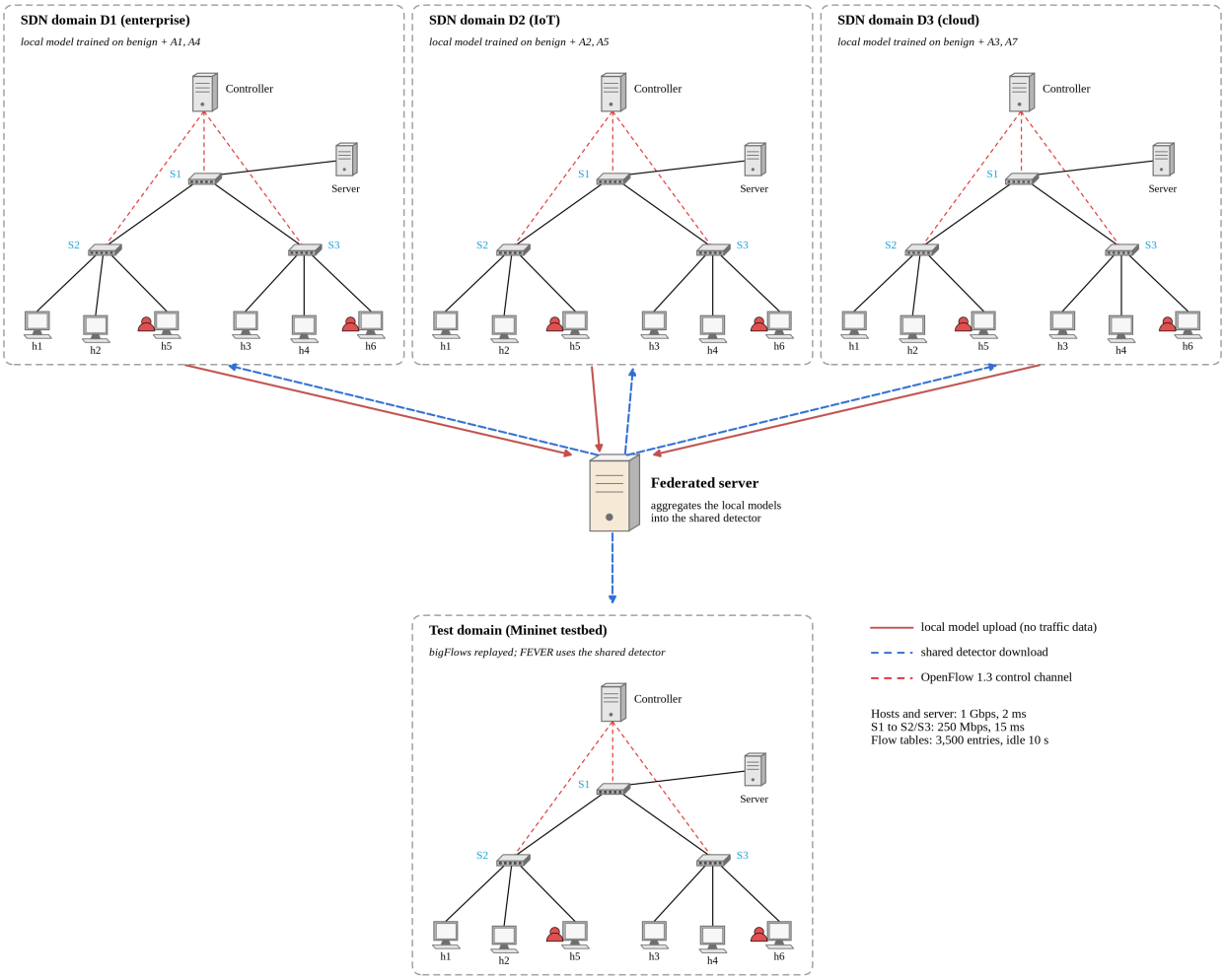


Fig. 5. Experiment topology: three SDN domains and the test domain.

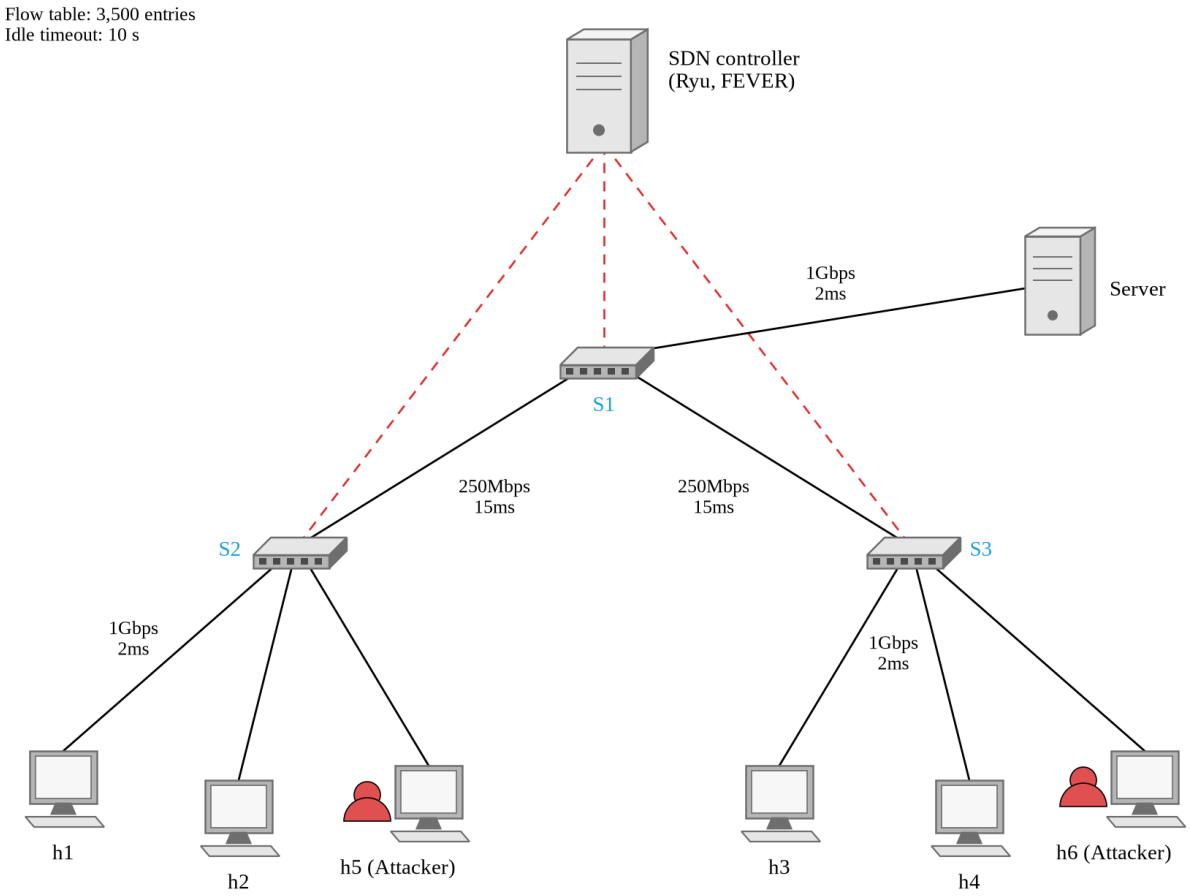
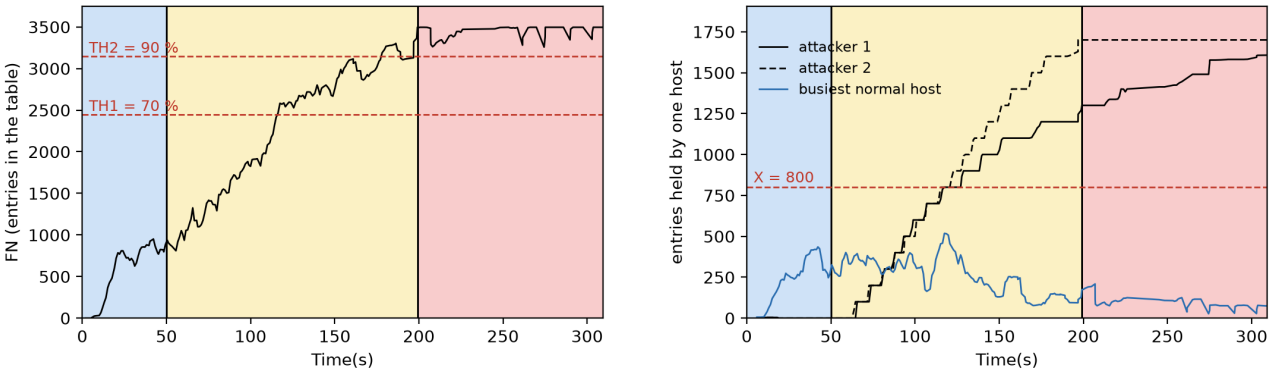


Fig. 6. Network topology for evaluation.

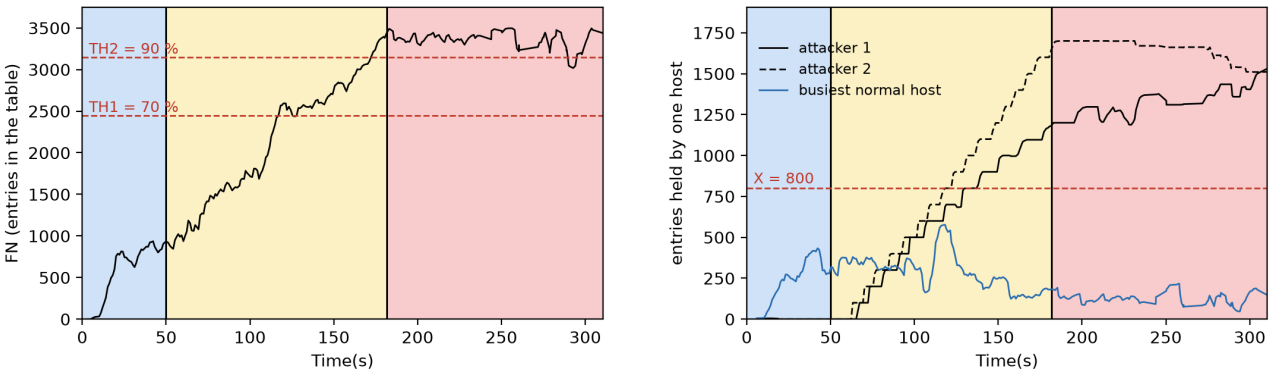
Flow table under attack without defense



(a) Distribution of FN before and after the attack.

(b) Entries per host before and after the attack.

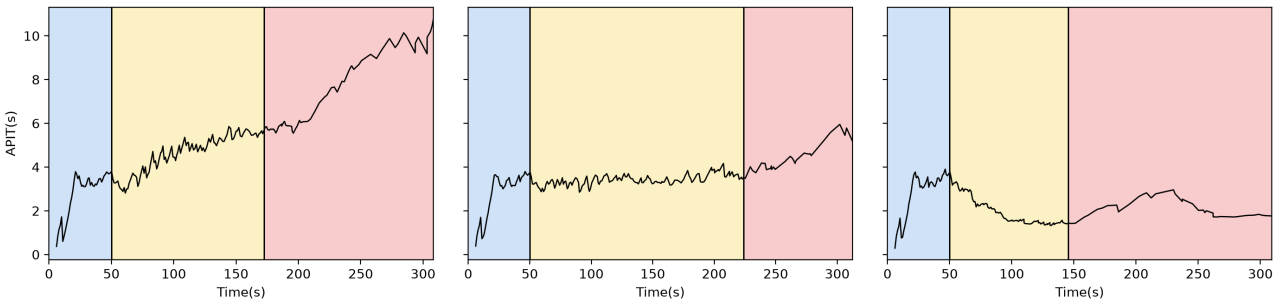
Fig. 7. Distribution of FN and of the entries held per host before and after the A2 attack.



(a) Distribution of FN before and after the attack.

(b) Entries per host before and after the attack.

Fig. 8. Distribution of FN and of the entries held per host before and after the A4 attack.



(a) APIT before and after A2.

(b) APIT before and after A4.

(c) APIT before and after E9P.

Fig. 9. APIT before and after the attacks.

Mitigation results on the testbed

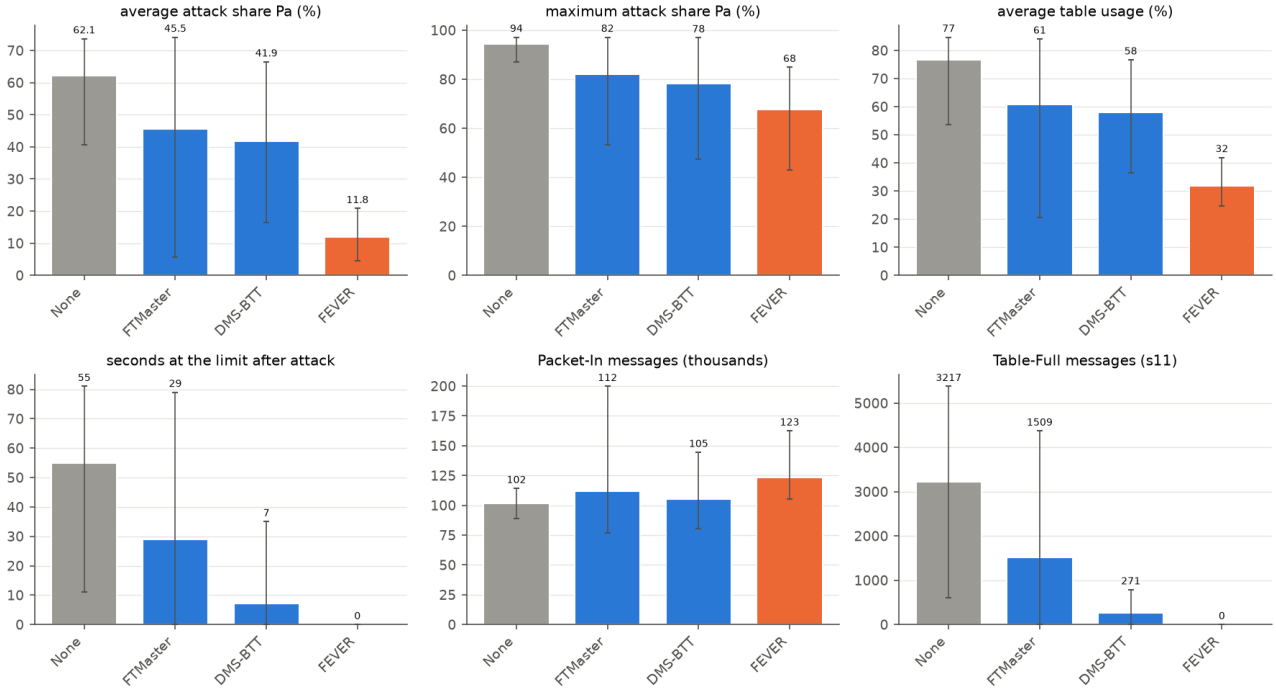


Fig. 10. Comparison of the defenses under A2, A4 and E9P (five runs each; bar: mean, whisker: range).

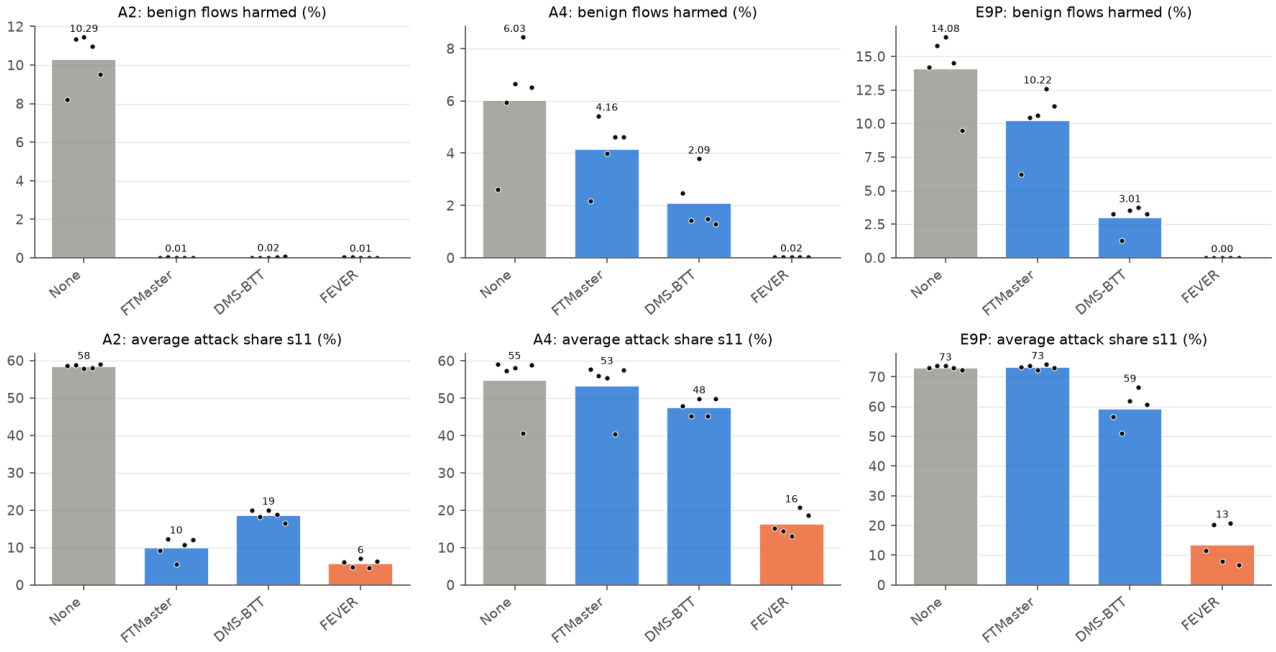


Fig. 11. Benign flows harmed and attack share under each attack (bar: mean of five runs; dot: one run).

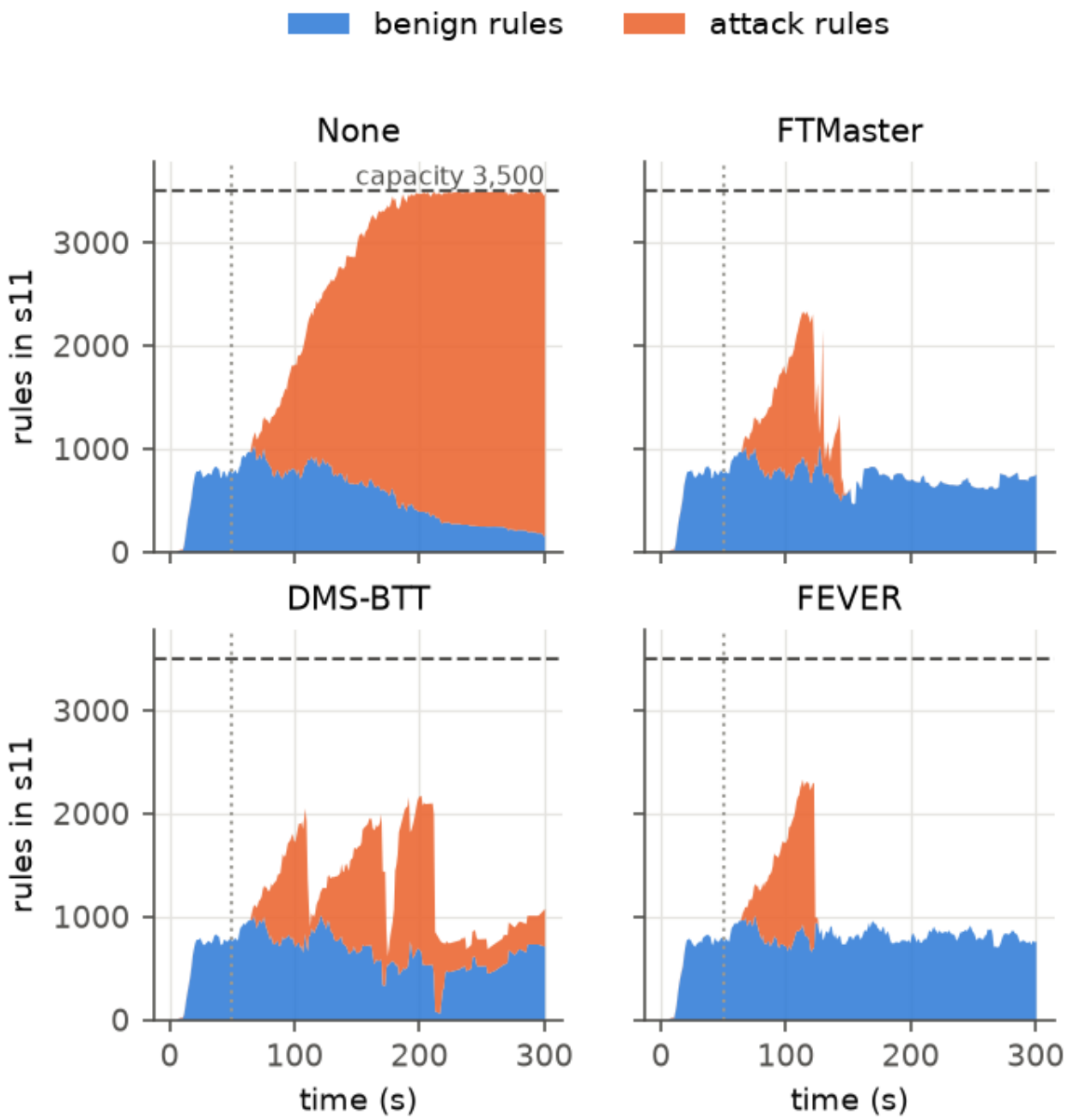


Fig. 12. Benign and attack rules in the flow table over time under A2.

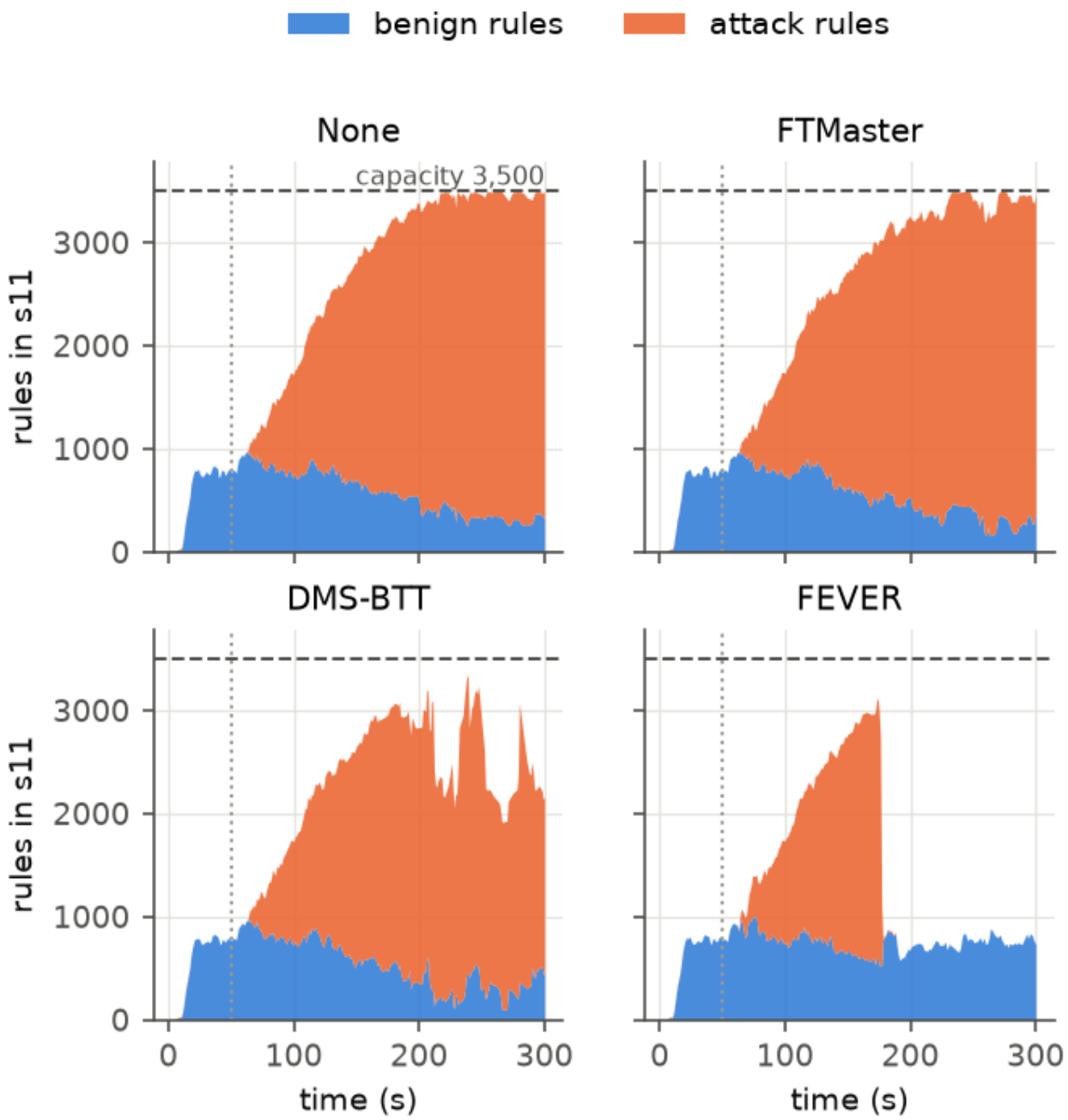


Fig. 13. Benign and attack rules in the flow table over time under A4.

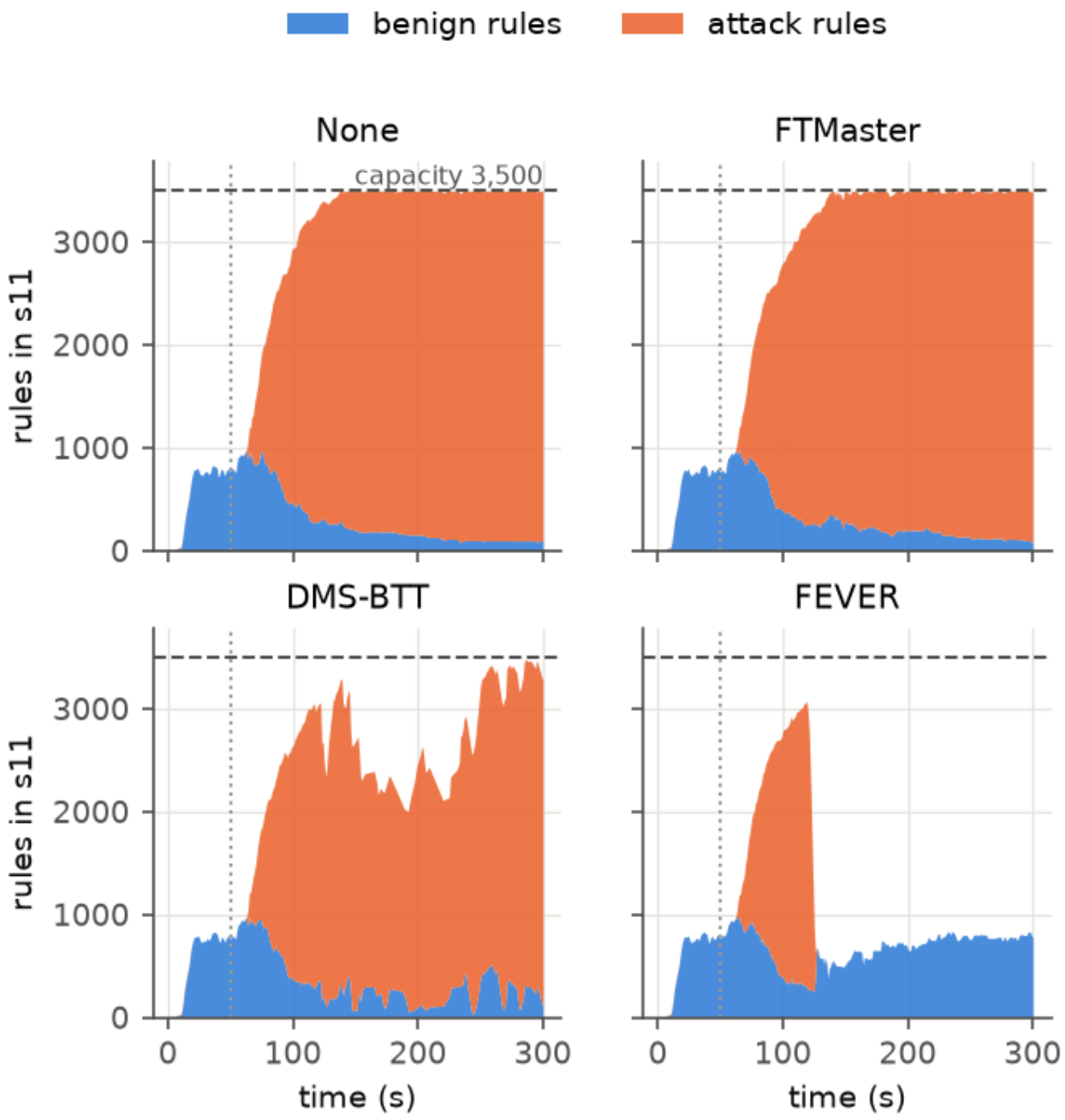


Fig. 14. Benign and attack rules in the flow table over time under E9P.

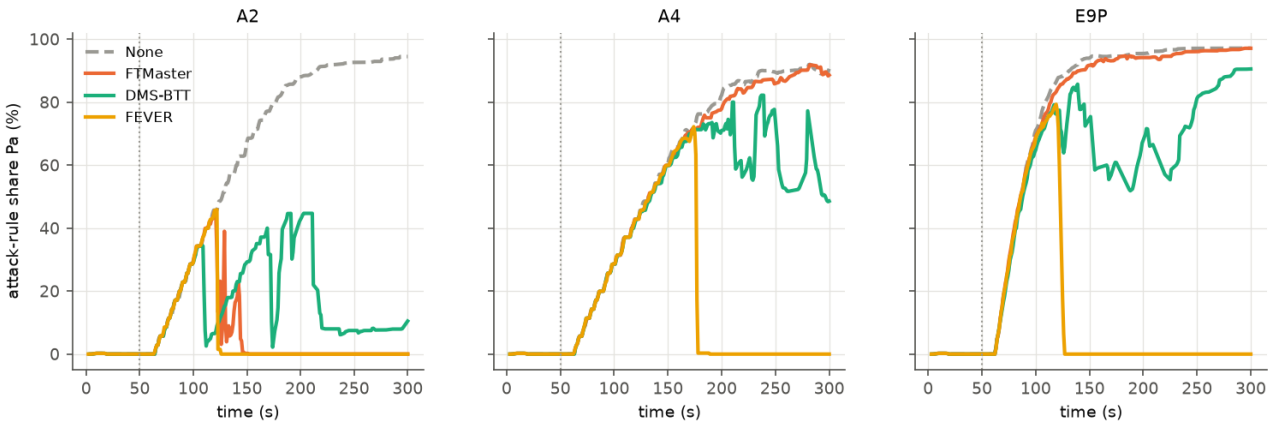
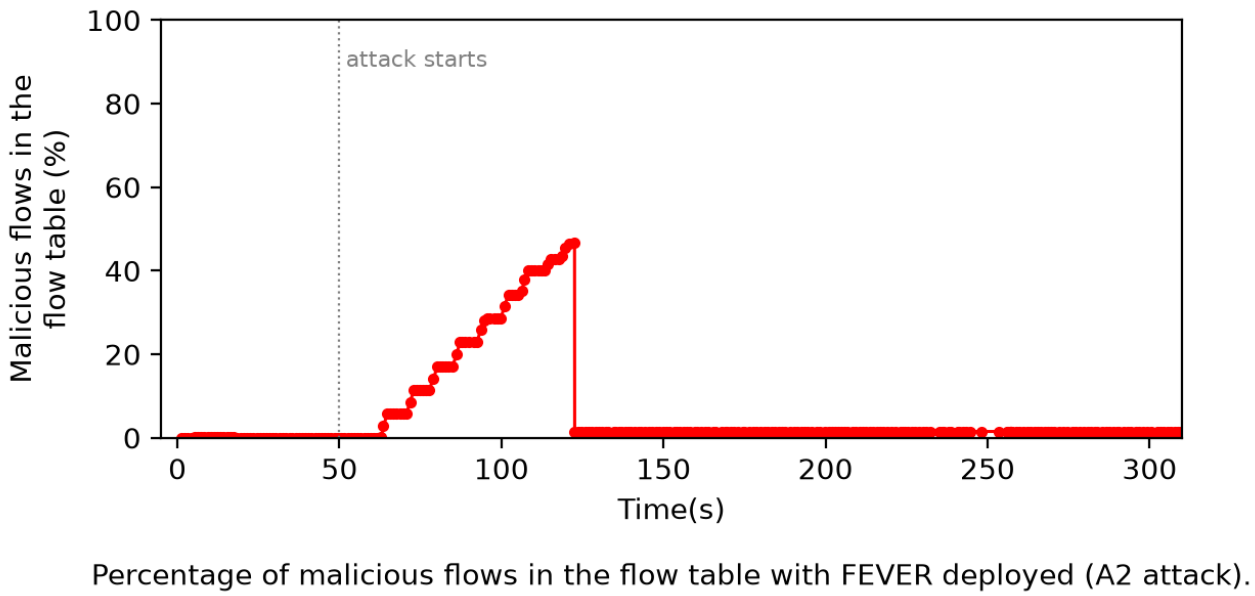


Fig. 15. Attack-rule share of the flow table over time.



Percentage of malicious flows in the flow table with FEVER deployed (A2 attack).

Fig. 16. Percentage of malicious flows in the flow table with FEVER under A2.

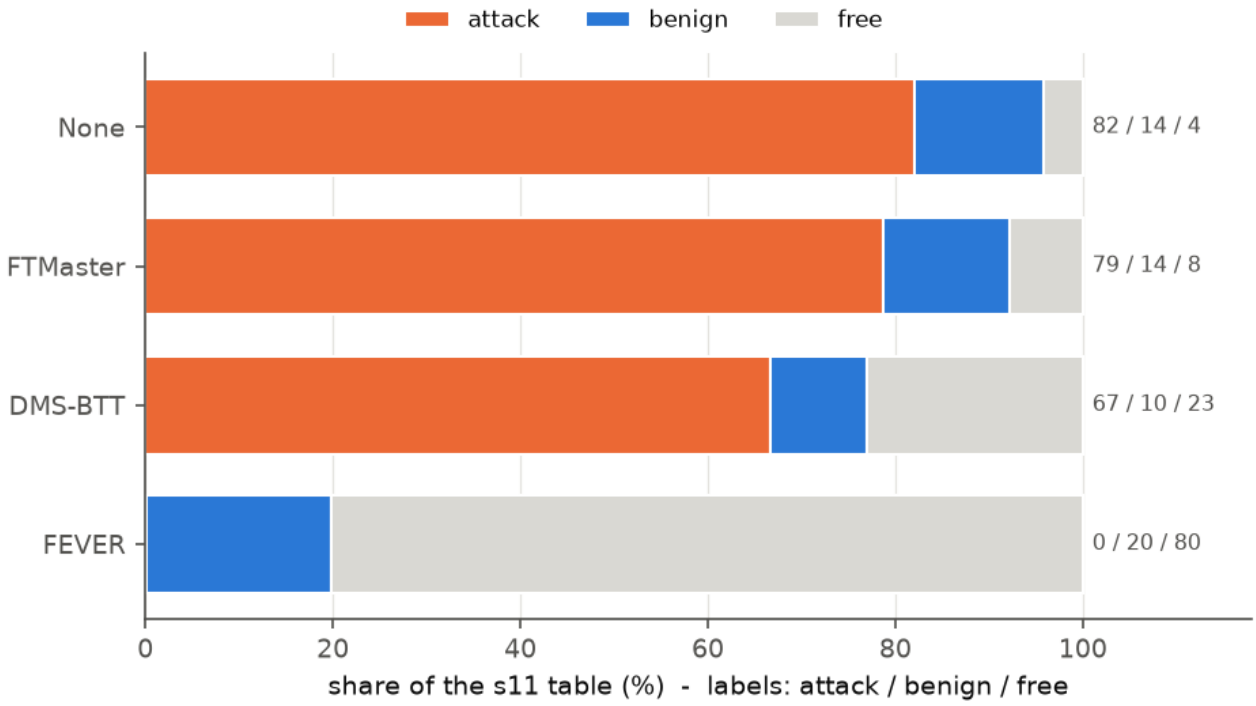
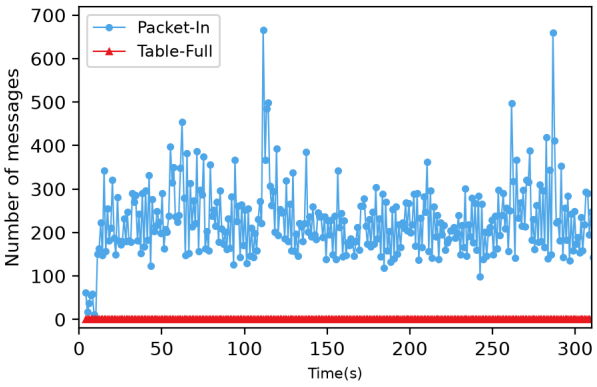
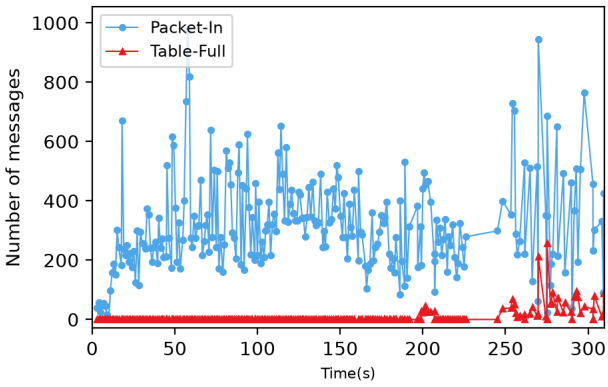


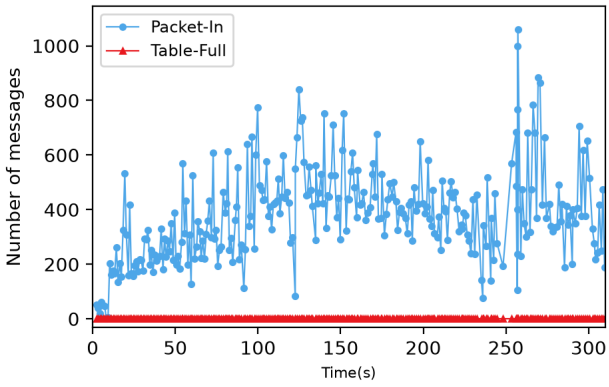
Fig. 17. Composition of the flow table 150 s into the A4 attack.



(a) No LFTO attack in the network.

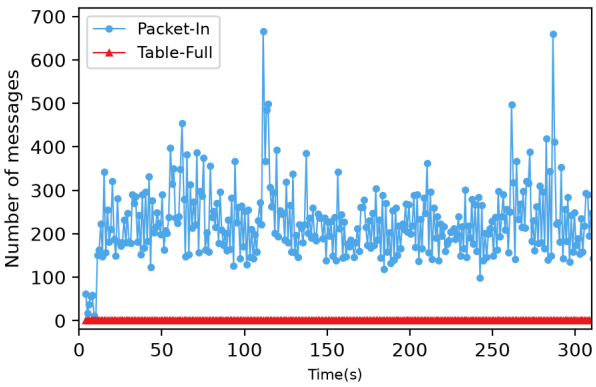


(b) LFTO attack launched and FEVER not deployed.

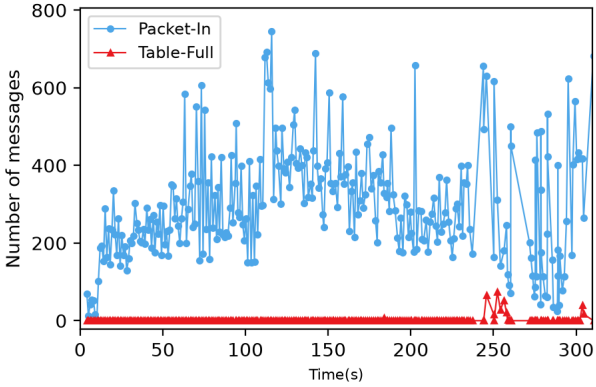


(c) LFTO attack launched and FEVER deployed.

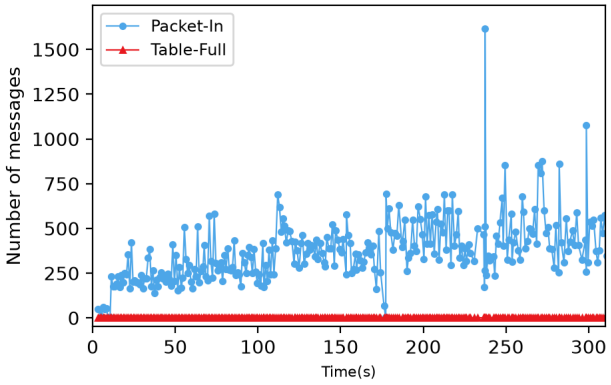
Fig. 18. The quantity of Packet-In and Table-Full messages under A2.



(a) No LFTO attack in the network.

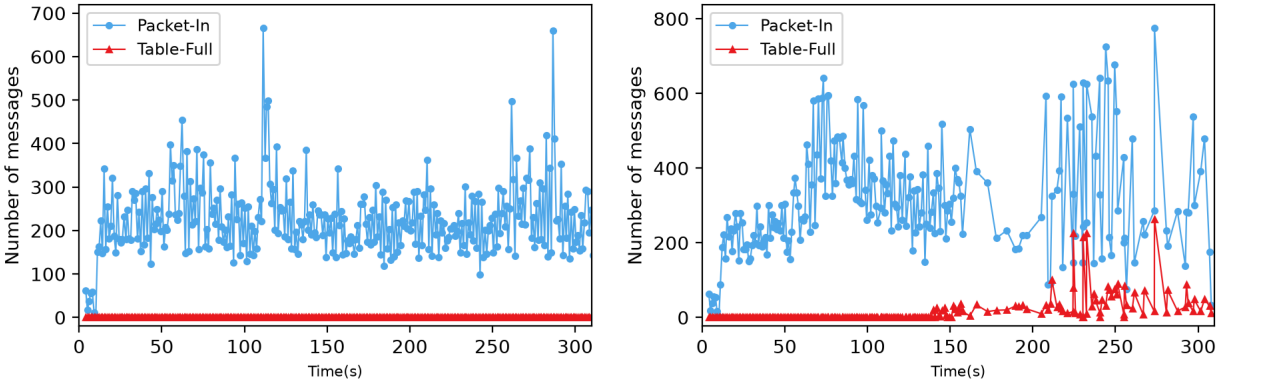


(b) LFTO attack launched and FEVER not deployed.



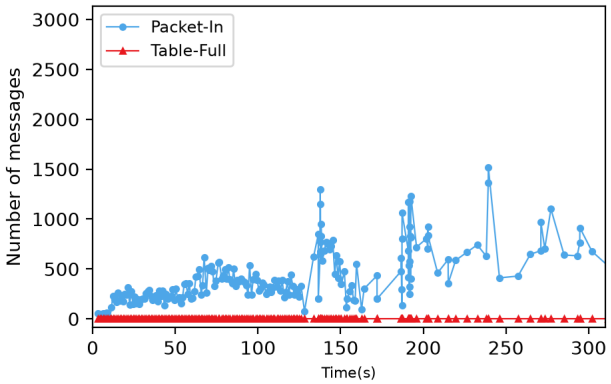
(c) LFTO attack launched and FEVER deployed.

Fig. 19. The quantity of Packet-In and Table-Full messages under A4.



(a) No LFTO attack in the network.

(b) LFTO attack launched and FEVER not deployed.



(c) LFTO attack launched and FEVER deployed.

Fig. 20. The quantity of Packet-In and Table-Full messages under E9P.

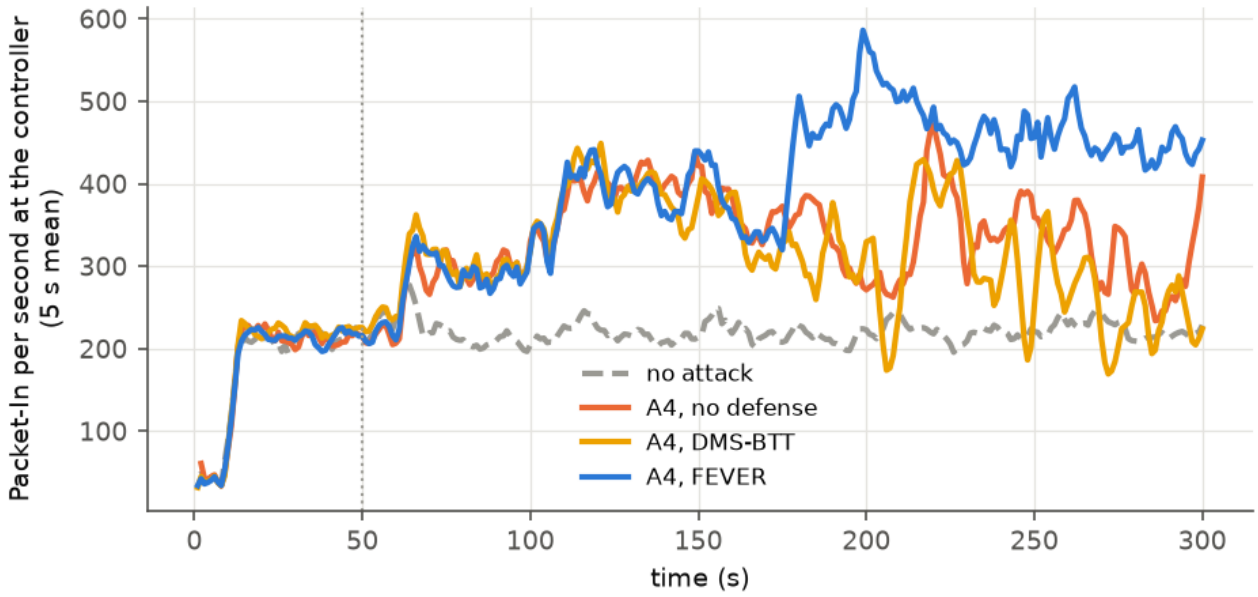


Fig. 21. Packet-In messages per second at the controller under A4.

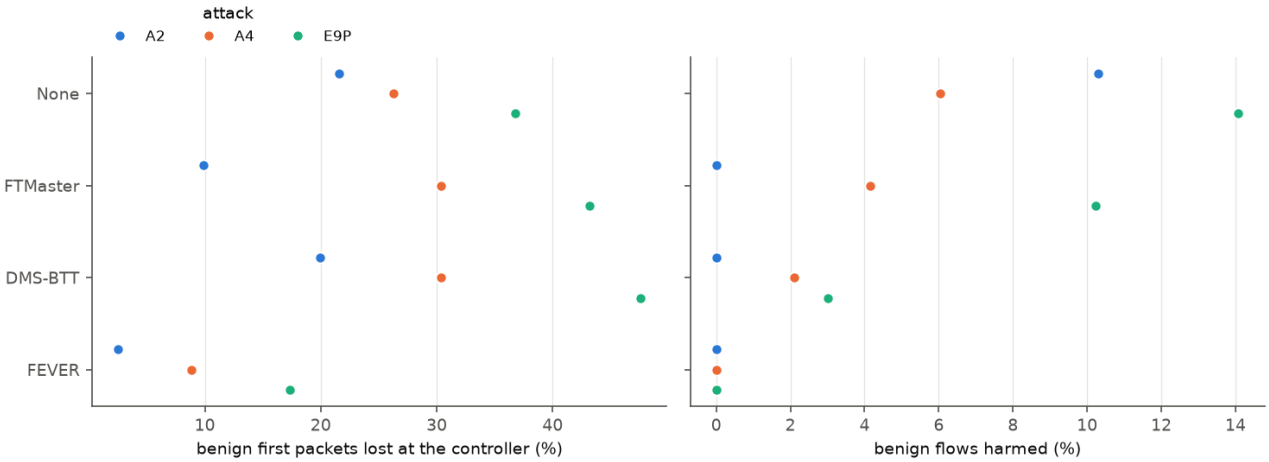


Fig. 22. Benign first packets lost at the controller and benign flows harmed.

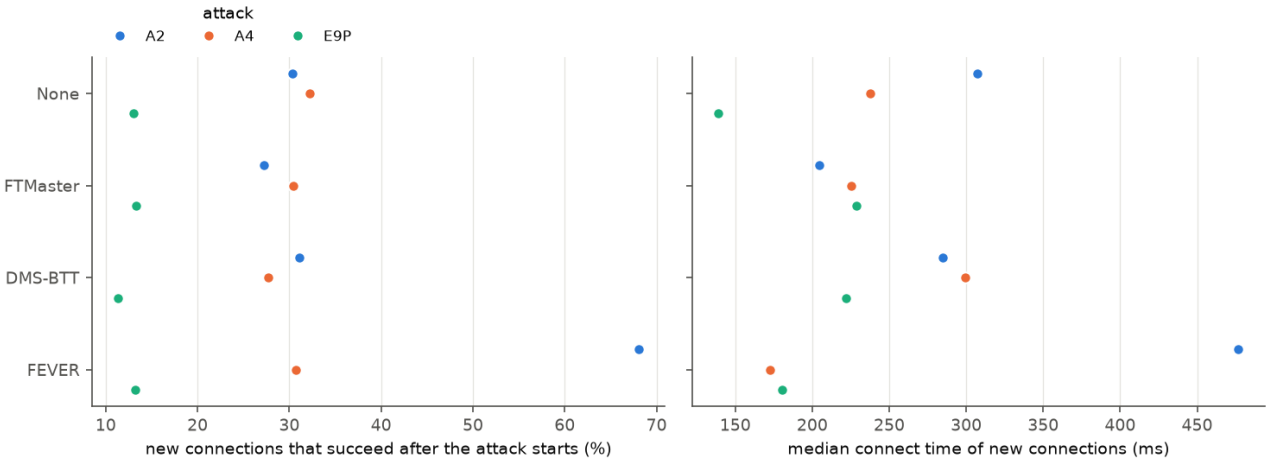


Fig. 23. New connections that succeed and their connection time.

Overhead

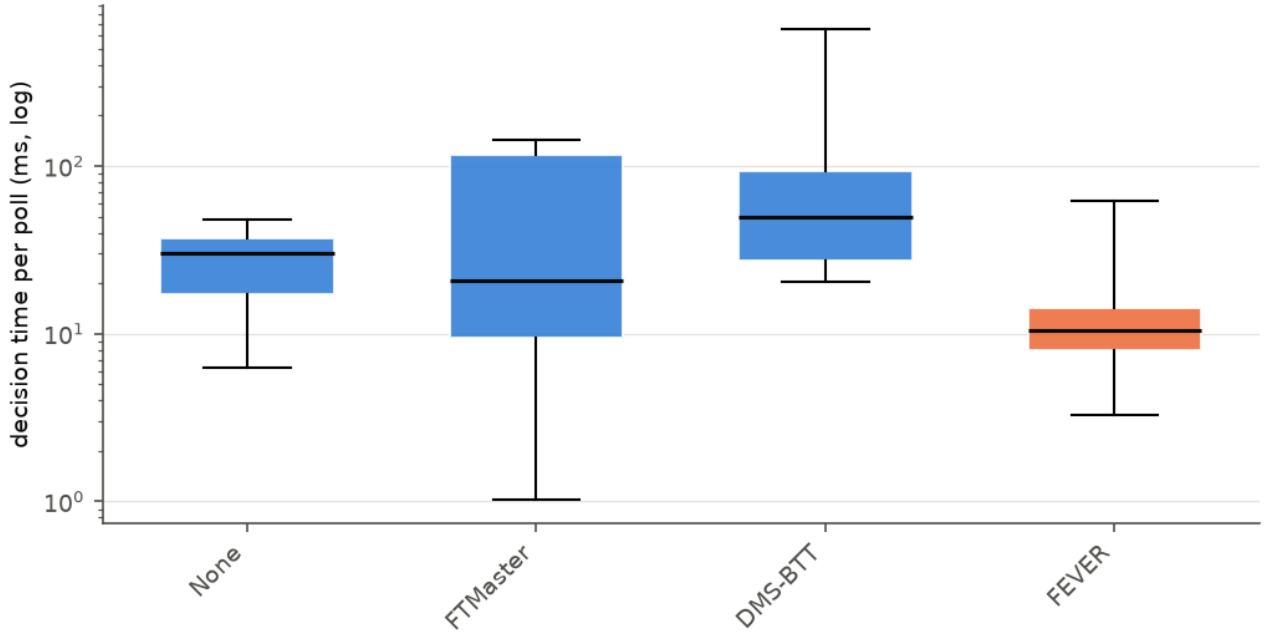


Fig. 24. Decision time per poll under attack.

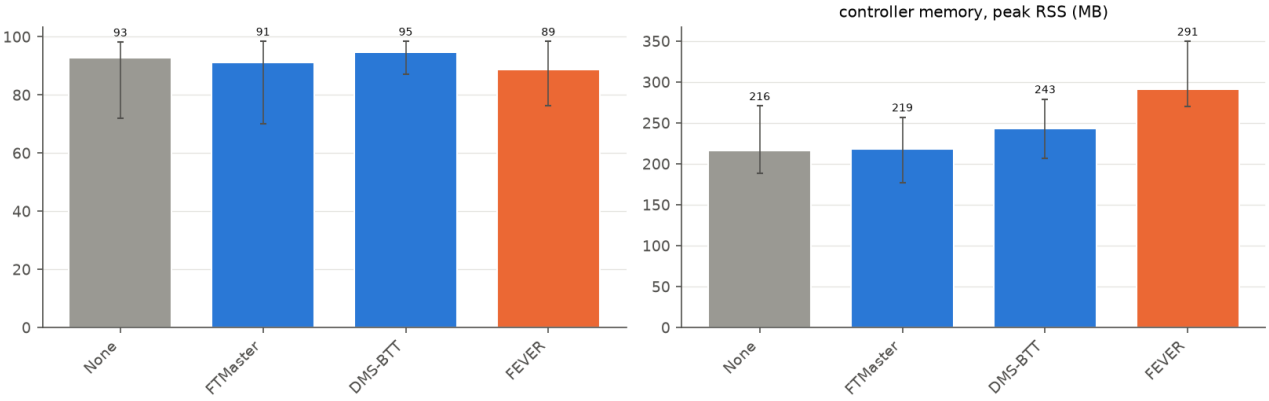


Fig. 25. Controller CPU and memory under attack.

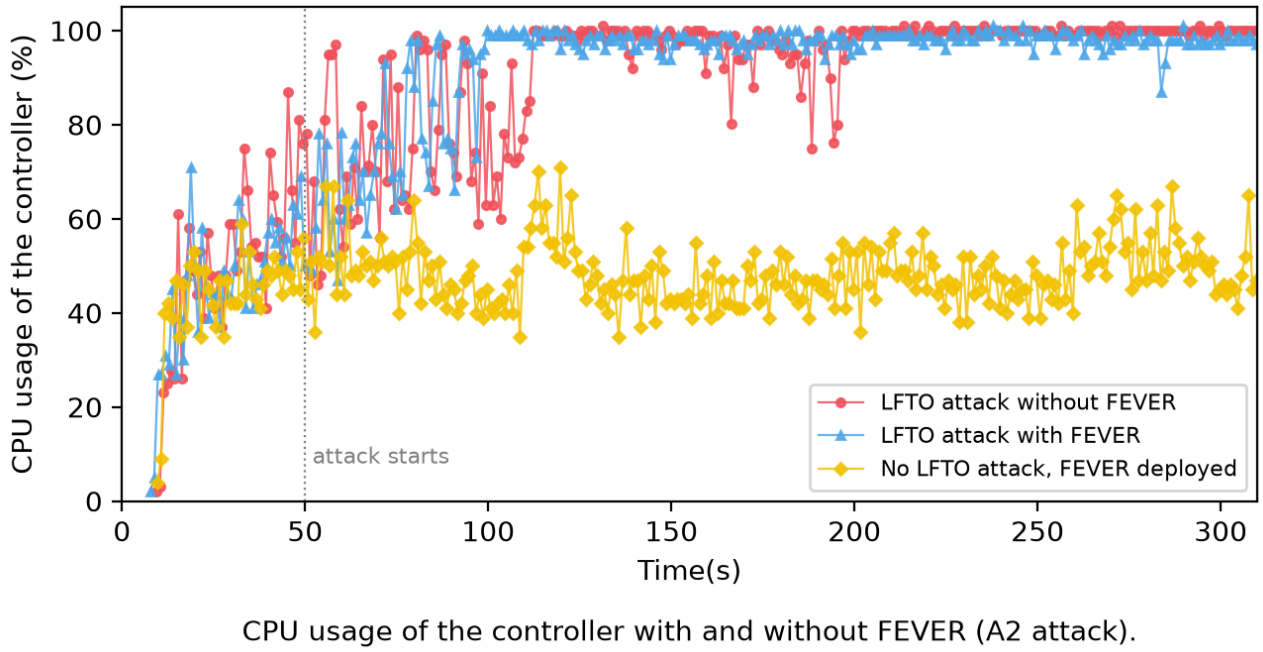


Fig. 26. CPU usage of the controller under A2.

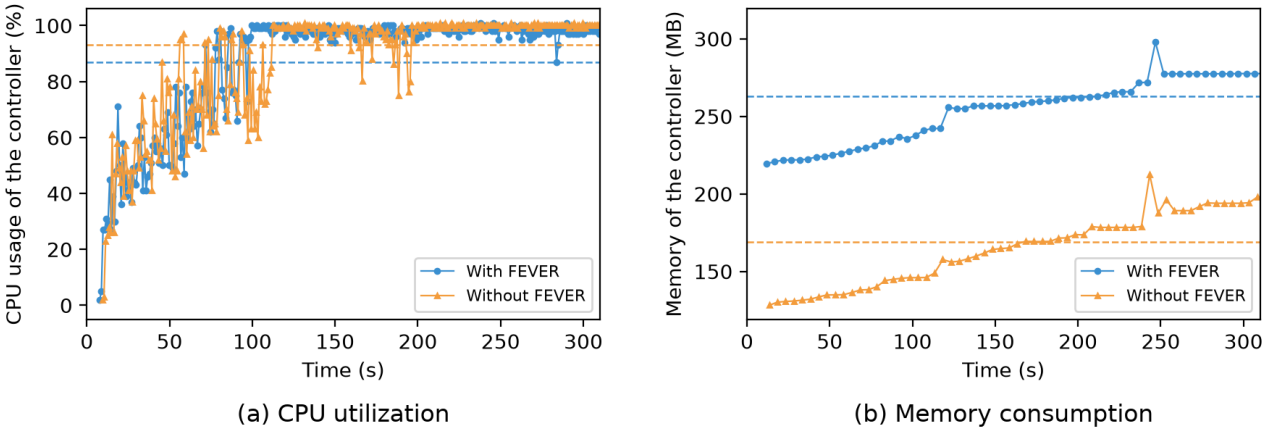


Fig. 27. System overhead of deploying FEVER.

Ablation

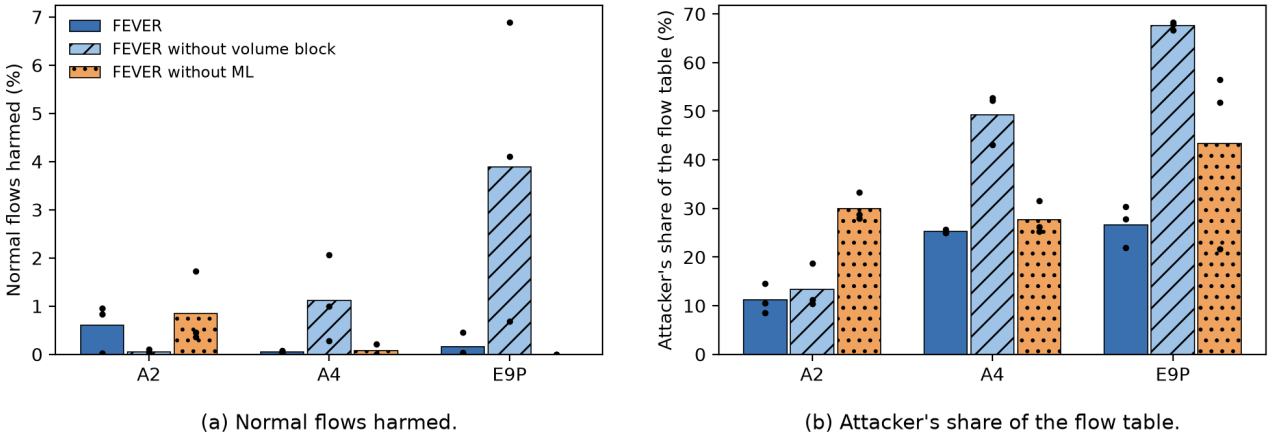


Fig. 28. Ablation of FEVER with the 60 s block: benign flows harmed and attack share under each attack.

Tables

Table 1
Comparison with published methods under LFTO attacks.

Method	Normal flows harmed (%)	Attacker's share of the table (%)	Attackers blocked	Innocent hosts blocked	New connections succeeding (%)	First packets lost (%)	Decision time (ms)	Extra memory (MB)
No defense	10.29	63.3	0 / 18	0	25.7	28.1	31	-
FTMaster	4.31	46.2	0 / 18	0	23.5	27.9	77	9
DMS-BTT	1.75	41.2	0 / 18	0	23.3	32.6	64	26
rDefender	1.83	45.0	0 / 18	0	20.8	36.4	26	31
FEVER	**0.02**	**11.1**	18 / 18	0	**33.0**	**9.9**	**10**	88

Table 2
Results under each attack.

Method	Normal flows harmed (%)			Attacker's share of the flow table (%)		
	A2	A4	E9P	A2	A4	E9P
No defense	10.33	5.07	15.46	58.4	58.1	73.5
FTMaster	0.01	3.86	9.07	9.0	56.3	73.1
DMS-BTT	**0.00**	2.56	2.69	19.4	47.6	56.5
rDefender	2.82	1.86	0.82	38.6	44.7	51.8
FEVER	0.02	**0.02**	**0.00**	**5.9**	**14.2**	**13.3**

Table 3
Accuracy, precision, recall and F1 score (%) of the detectors.

Detector	Attacks seen in training				Attacks seen only by another controller				New attack seen by no controller			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
FTOP	96.94	99.79	92.02	95.02	92.59	84.33	79.43	81.16	81.18	35.50	9.49	14.74
FTMaster	97.07	99.79	92.37	95.30	93.34	84.41	81.29	82.20	82.37	35.64	15.00	20.65
FloRa	99.01	**99.82**	97.50	98.62	93.65	84.30	82.19	82.68	84.51	35.48	24.90	28.46
DMS-BTT	97.03	99.78	92.27	95.22	93.45	84.29	81.62	82.38	82.21	35.36	14.28	19.94
FEVER	**99.61**	99.75	**99.19**	**99.46**	**99.43**	**99.75**	**98.60**	**99.15**	**89.48**	**98.99**	**50.98**	**66.81**

Table 4
Contribution of each part of FEVER.

Variant	Normal flows harmed (%)			Attacker's share of the flow table (%)	Attackers blocked	Time to block under A2 (s)	Innocent hosts blocked
	A2	A4	E9P				
FEVER	**0.02**	**0.02**	**0.00**	**11.1**	18 / 18	72	0
FEVER with a 60 s block	0.61	0.05	0.16	21.0	18 / 18	**71**	0
60 s block, without volume block	0.05	1.11	3.89	43.4	6 / 18	**71**	0
60 s block, without ML	0.84	0.07	**0.00**	33.7	16 / 18	130	0

Table 1
System overhead of each method.

Method	CPU usage (%)			Memory (MB)		Time per decision (ms)
	No attack	Increase over no defense	Under attack	Peak	Increase over no defense	
None	48.2	-	92.9	216	-	30
FTMaster	48.6	+0.4	91.2	219	+2	72
DMS-BTT	52.4	+4.3	94.6	243	+27	65
FEVER	49.2	+1.1	88.9	291	+75	11